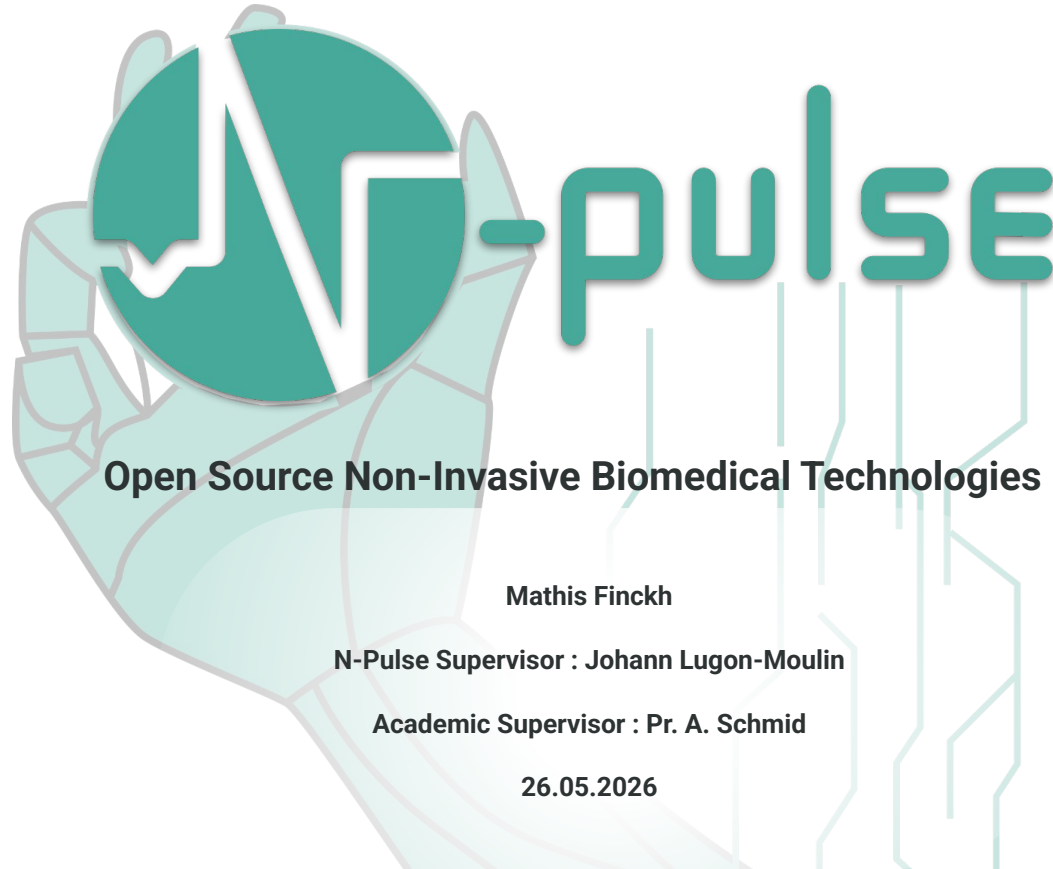


Semester Project Final Presentation

1

Comprehensive Design of the Safety System of a Prosthesis



Open Source Non-Invasive Biomedical Technologies

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N-Pulse Supervisor : Johann Lugon-Moulin

Academic Supervisor : Pr. A. Schmid

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Overview

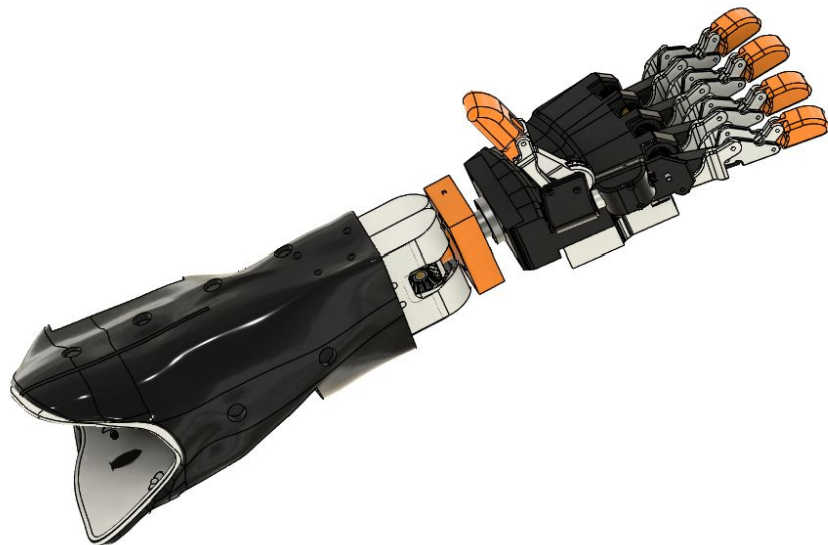
1. **Context & Introduction:** N-Pulse mission and the need for a software-controlled prosthetic safety system.
2. **Hardware & Component Selection:** Overview of the chosen sensors (thermal, humidity, electrical, mechanical) and the MCU power disconnect.
3. **Software :** Explanation of code logic
4. **Engineering Justifications:** Mathematical models validating thermal responses, capacitance, electromagnetic noise, and shock thresholds.
5. **Conclusion & Next Steps:** Final prototype results, physical limitations, and plans for future testing.



1. Context & Introduction



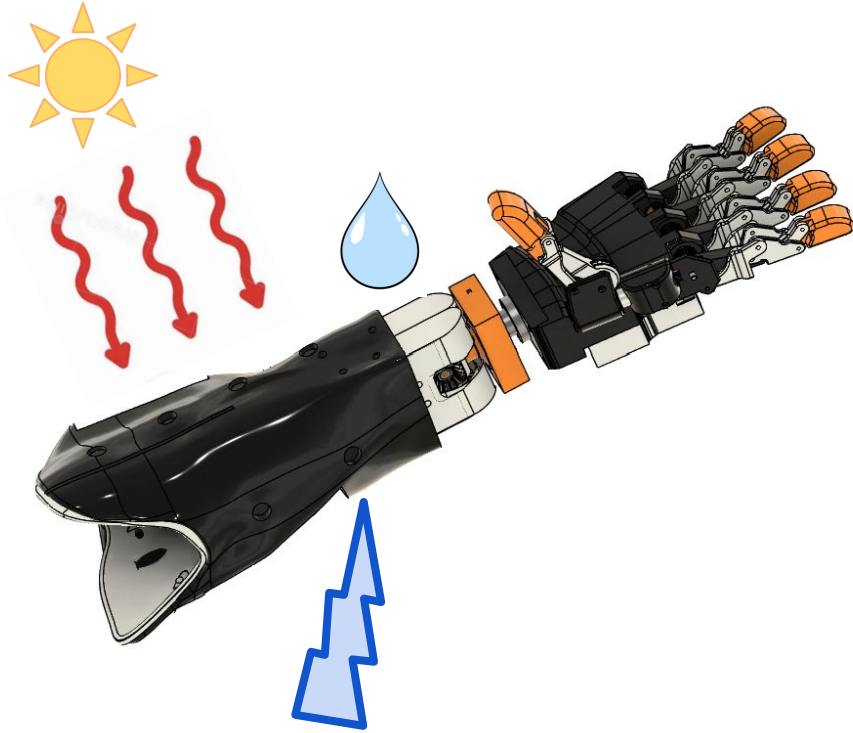
N-Pulse



- EPFL MAKE project
- Build cutting edge, open-source, accessible, non-invasive biomedical technologies
- BMI : EMG, EEG, and computer vision controls
- Prosthesis : Mechanical, electrical and software design of the arm



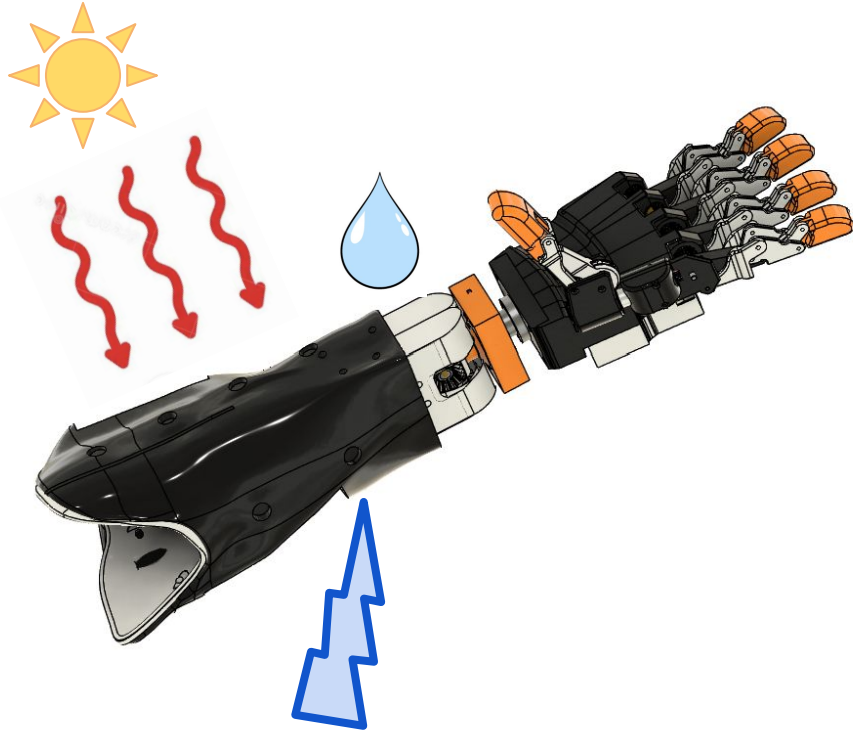
Current Safety Limitations



- Currently fuse on power board as main protection
- Risks :
 - Over Currents
 - Overheating in battery and motors
 - High temperature of ambient air for
 - electronics
 - patient skin contact
 - High relative humidity
 - High impacts



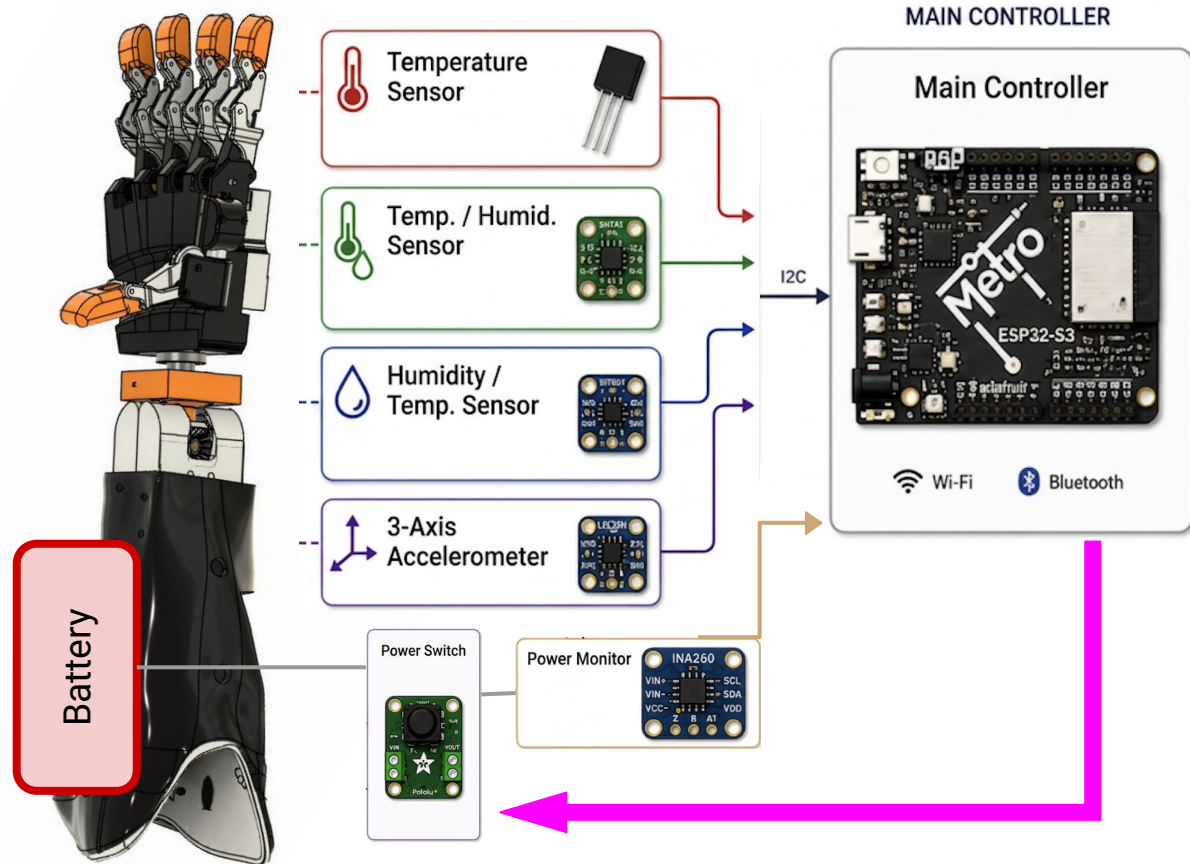
Project goal



- Create software-controlled battery disconnect system
- Monitors sensors for each risk
- Provide diagnostic data for troubleshooting
- Live visualization
- Data storage and analysis
- Electrically controlled battery disconnect



Solution Visualisation



2. Hardware & Component Selection

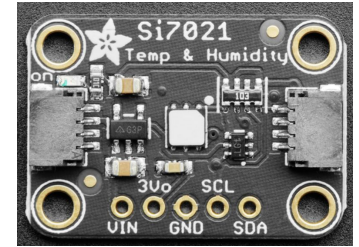
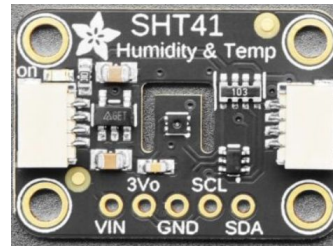


Component wise limits

Component	Range	Source
Battery	0 to 60°C (45 from QMS	QMS (manufacturer recommendation)
Motor casing	-10 to 80 °C	Datasheet
Electronics	-40 to 85 °C	Different sources, general electronics safety
Socket	max 41°C on skin	Medical device safety
Shock	16g	Chosen
Humidity	70% very safe	Different sources, general electronics safety



Relative Humidity Sensor



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Need :

- Compact enough to fit inside the prosthetic enclosure
- Able to monitor humidity (up to 100%) as well as ambient temperature
- Easy to integrate into the existing I2C-based system

Chosen sensors:

Adafruit SHT41 (inside arm, near electronics)

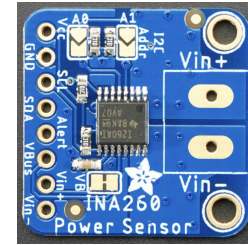
- Measures relative humidity and temperature with high precision.
- Very low power consumption.
- Suitable for enclosed spaces where moisture buildup may matter.

Adafruit Si7021 (Socket)

- Idem with protective PTFE filter (for more exposed usage at socket)



Current Sensor



Need :

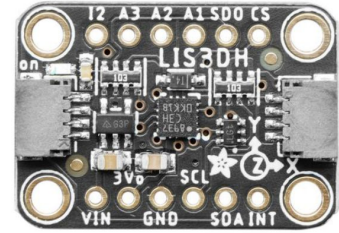
- Must monitor battery current in real time.
- Should help **detect overcurrent**, short circuits, or abnormal power draw : **Max 5.3 A from motors**
- Can handle **>2x max current (10.6A)**
- Easy to use with the main controller during prototyping.

Chosen sensor: Adafruit INA260

- Measures **current, voltage, and power** in one device.
- Works over **I2C**
- **Programmable alert capability.**
- Can handle **15A continuously**
- Designed for straightforward integration into battery monitoring.



Accelerometer



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Need :

- Must detect sudden shocks, falls, or abnormal impacts (**>10g**)
- Should be sensitive enough to identify dangerous mechanical events.
- Compact and easy to mount on the prosthetic structure.

Chosen sensor: Adafruit LIS3DH

- 3-axis accelerometer with a **± 16 g** full-scale range.
- Includes **hardware interrupts** for shock or free-fall events.
- Small, low-power, and well suited to embedded safety detection.
- **10'000 g high shock survivability**



Thermal Sensor

Need :

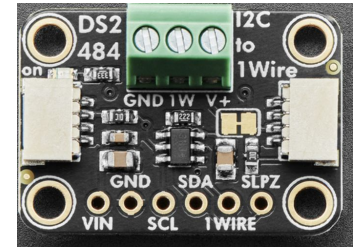
- Small enough to be attached to motors, battery, and socket
- Can withstand and sense **>85°** (max. allowed motor casing temp)
- Easy to use for prototyping and implementation

Chosen sensor : DS18B20

- Measures Temperatures from **-55°C to +125°C**
- **1-Wire®** Interface (no calibration)
- Small : < 1cm



1-Wire to I2C Adapter



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Need :

- Must allow the 1-Wire thermal sensors to communicate with the main controller
- Should simplify wiring to only have I2C cables passing by the wrist

Chosen component : Adafruit DS2484

- Converts 1-Wire to I2C communication.
- Makes it easier to integrate multiple DS18B20 sensors.
- Reduces software and wiring complexity.



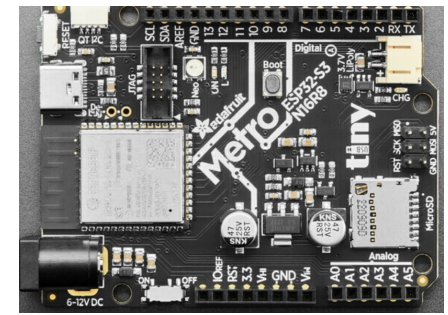
Microcontroller

Need :

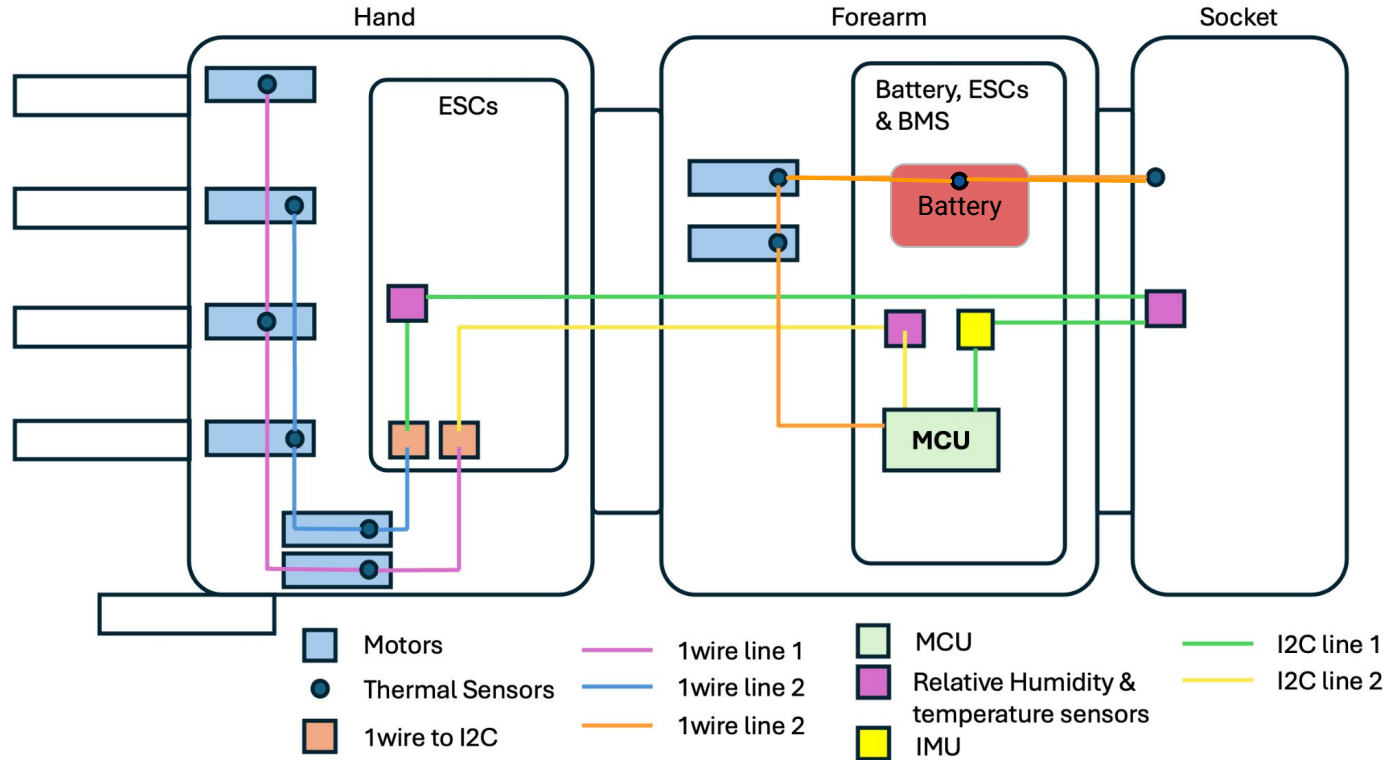
- Handle **2 separate I2C lines**
- Enough performance for future expansion and diagnostics.
- **Extra pins** for hardware interrupt
- Possibility for wireless communication
- **MicroSD** slot for data storage in case of failure and power cut.
- Low consumption

Chosen MCU: Adafruit Metro ESP32-S3

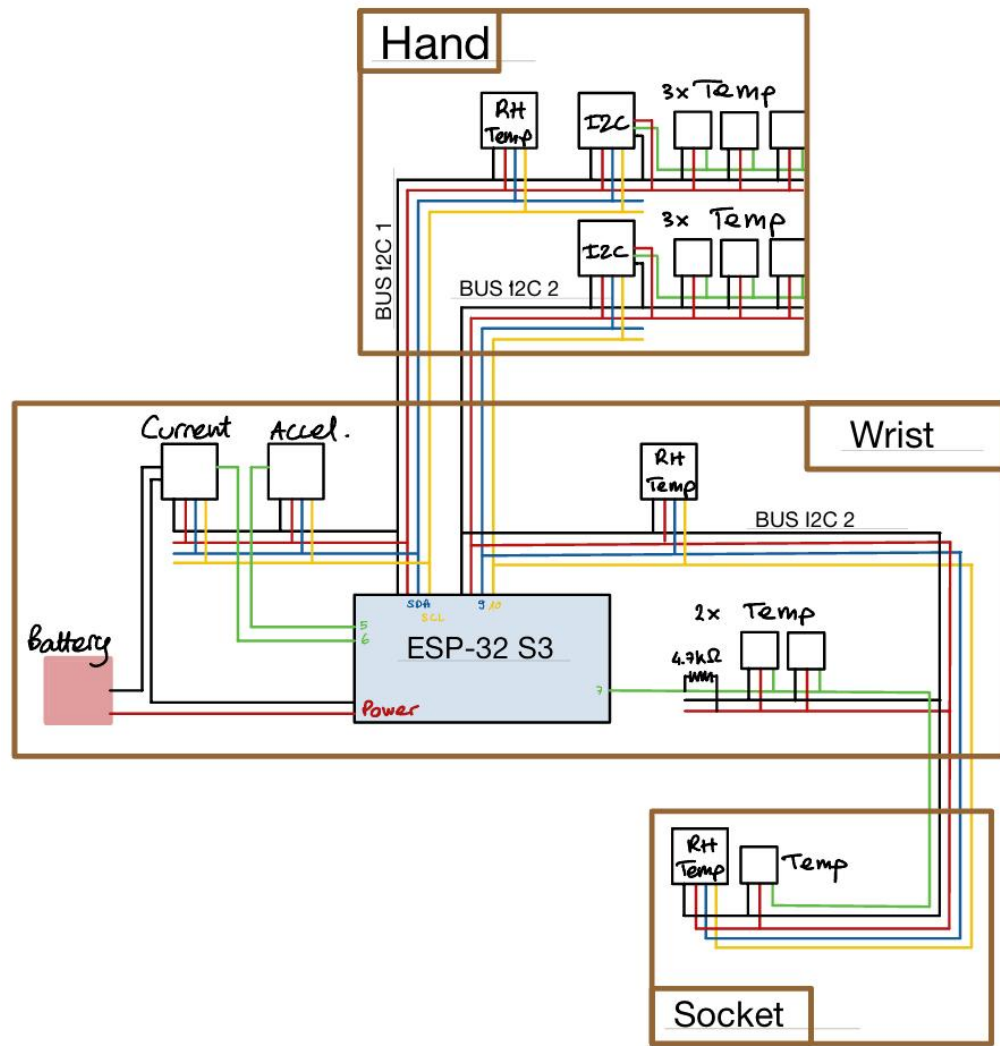
- Powerful MCU with wireless capability.
- Supports both I2C and 1-Wire-related system architecture.
- Good fit for a safety-oriented embedded prototype.
- Many pins
- MicroSD slot
- 60-80 mA in use
- Low Power deep sleep mode (100uA)
- Stemma QT connector for quick prototyping



Bus Architecture



Bus Architecture



3. Software



Embedded Safety Logic – Pseudocode

Initialize sensors and interrupts

Loop:

```
    check interrupt-based faults
    If a critical fault is confirmed:
        request power cut

    update temperature measurements

    every logging period:
        read all sensors
        evaluate thresholds
        log data

    update warning LED
```



3. Engineering Justifications & Modeling



I2C Capacitance Budgeting

$$C_b < \frac{t_{rise}}{0.8473 \cdot R_p}$$

- Sources of capacitance : pins, connectors, cables, perfboard stray capacitance, sensor themselves
- Parallel pull-ups on sensor breakout boards (10kΩ each), sets line resistance and therefore the real capacitance limit
- Empirical confirmation: communication worked on the prototype
- Could theoretically allow for ~10 sensors on 1 line before hitting minimum resistance limit

Source	Qty / Length	Contribution [pF]
IC Pins	5 devices	50
Connectors	8 units	24
Cables	0.4 meters	20
Perfboard	10 cm	10
Total C_b		104

Source	Qty / Length	Contribution [pF]
IC Pins	4 devices	40
Connectors	8 units	24
Cables	0.6 meters	30
Perfboard	10 cm	10
Total C_b		104

Table 1: Capacitance budget for I²C Line 1 ($R_p = 2.5 \text{ k}\Omega$, $C_{b,max} = 141.6 \text{ pF}$). Table 2: Capacitance budget for I²C Line 2 ($R_p = 3.33 \text{ k}\Omega$, $C_{b,max} = 106.2 \text{ pF}$).



Electromagnetic Noise & Crosstalk

- Casing damps most of direct noise (Faraday cage), most risk between cables (Crosstalk, due to back flow of commutation and PWN noise)
- Modeled geometry: motor and I2C cables parallel, 5 mm apart, over 50 mm
- Coupling mechanisms: capacitive and inductive
- Noise estimates
 - **Stall case: about 2.13 mV**
 - **Nominal operation: about 1.63 mV**
- I2C Low-level margin: 330 mV → **Safe**
- Checked experimentally by routing cables close together and looking for abnormal readings → **No disturbance**



Shock Detection Thresholds

- **Everyday activities** can reach up to about **10 G** at wrist (running, jumping)
- Literature values for harsh events:
 - punch: about **80–230 G**
 - fall on outstretched hand: about **80–130 G**
- Simple drop estimate: ~ 177 G (1m fall, 4ms impact)
- Selected threshold: **16 G**, matching the LIS3DH full-scale range



- Gray zone of exact implementation, will require testing (hard knocks ?)



System Reaction Time

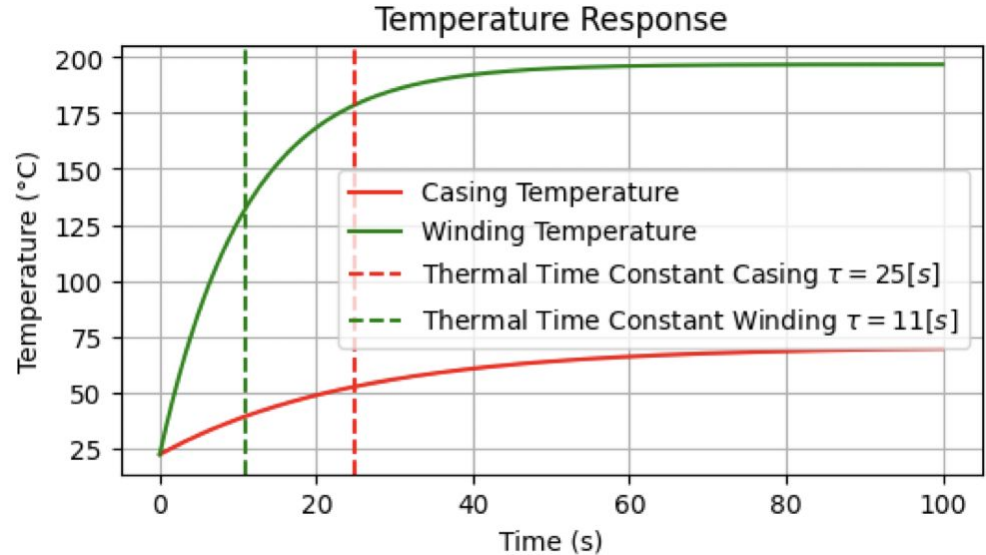
- Main loop idle execution time: about 5–15 μs (240MHz)
- Current fault reaction time: about 1.3 ms, dominated by INA260 conversion time
- Shock fault reaction time: about 20 ms, dominated by the LIS3DH duration filter at 200 Hz and 4 samples
- Design interpretation:
 - fast electrical faults are cut almost immediately
 - shock detection intentionally requires persistence to avoid false positives (doesn't require instant reaction either way)
- The architecture is hybrid: fast interrupt-based protection for urgent events, slower periodic monitoring for slow variables



Poling Frequency & Thermal Response

- Current and shock interrupt-based
- Temperature only variable potentially limited by polling frequency
- Under **stall**, the winding reaches 100°C in about **6.5 s**, while the casing is still only about **33°C**
- 25 s much slower than 0.5 s poling time

→ casing temperature is useful for **user/structure protection and diagnostics**, but not for direct winding protection under stall



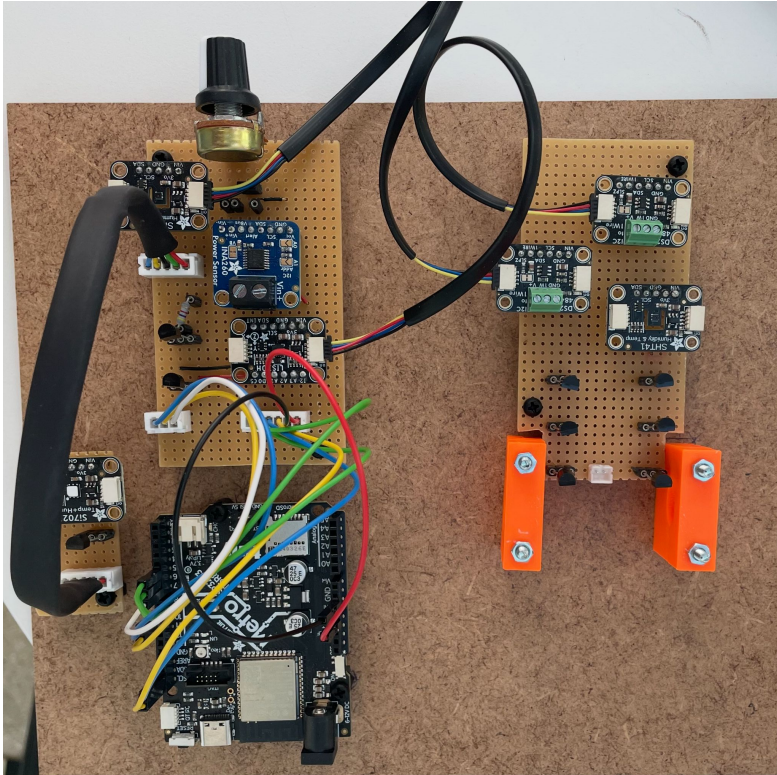
*Thermal resistance : 63% of the steady state temperature



5. Conclusion and Results



Final Results & System Successes

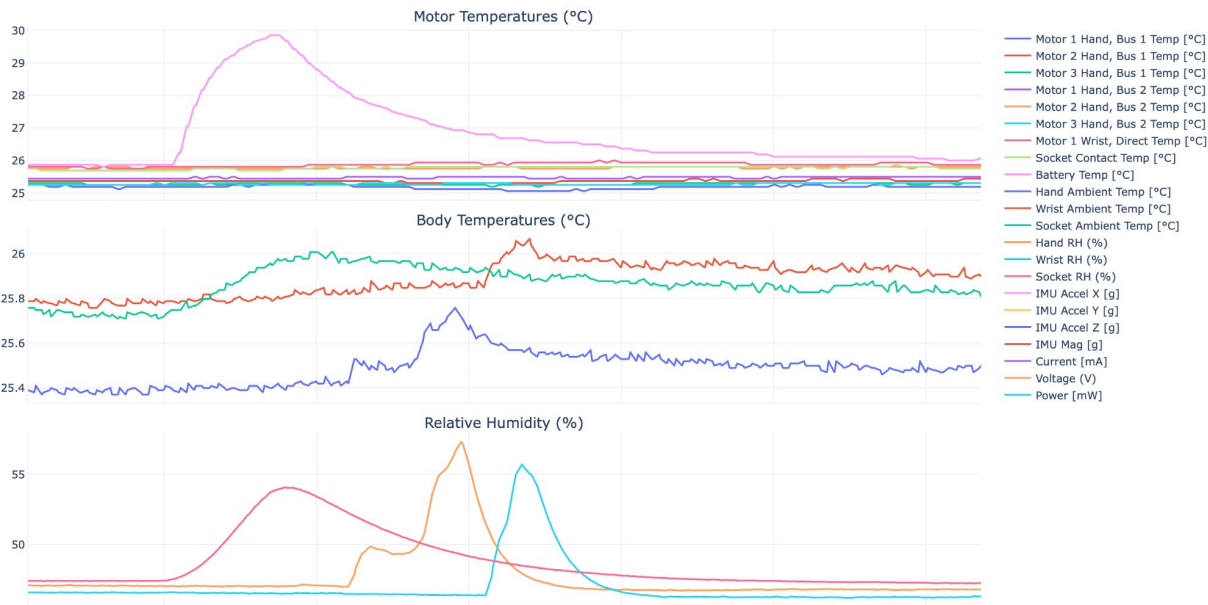


- I2C & 1Wire communication successful
- Reacts to hardware interrupt
- Flags passed thresholds
- Complete build of test bench



Final Results & System Successes

Prosthetic Sensor Live Telemetry



Graphical User Interface (GUI)

- Live plotting of measured metrics
- Visualisation of flags
- Visualisation of power cut
- Possibility to remove some metrics from the view



Known Limitations

- Power button & MicroSD logging not implemented or tested since hardware never arrived
- Very simple prototype, needs to be completely reworked to be implemented or tested with the arm
- Likely false positives (power cut) in case of a hard knock, requires calibration on final implementation (worst cast sensor with wider g range)



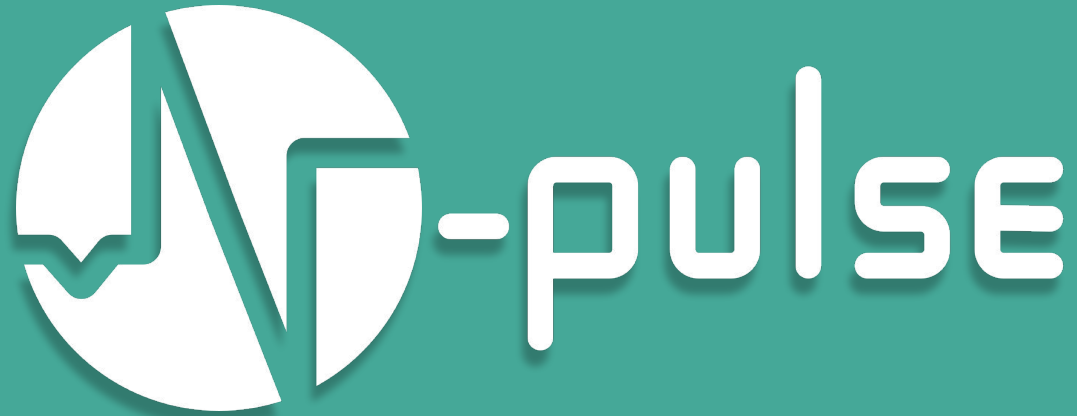
Next Steps

- Go further on the prototype
 - Implement Power cut switch and data storage
 - Testing with motor under load (Electromagnetic noise, overheating ?)
 - Prediction using gradient
 - Implement on the real prosthetic to validate before redesign
- Redesign
 - PCB for the hand that regroups all sensors
 - PCB for MCU and wrist sensors



Questions ?





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